where they are in the Universe and regardless of how fast they are moving.

- Maxwell's laws of electromagnetism must take on the same mathematical form in all reference frames. They failed to do so, when physics was built on Newtonian foundations (magnetic field lines could have ends in some frames but not in others), and this failure was deeply disturbing to Lorentz, Poincaré, Larmor, and Einstein.

In Einstein's view it was utterly unacceptable that the laws were simple and beautiful in one frame, that of the aether, but complex and ugly in all frames that moved relative to the aether. By reconstructing the foundations of physics, Einstein enabled Maxwell's laws to take on one and the same simple, beautiful form (for example, "magnetic field lines never ever have any ends") in each and every inertial reference frame—in accord with his principle of relativity.

The principle of relativity is actually a *metaprinciple* in the sense that it is not itself a law of physics, but instead is a pattern or rule which (Einstein asserted) must be obeyed by *all* laws of physics, no matter what those laws might be, no matter whether they are laws governing electricity and magnetism, or atoms and molecules, or steam engines and sports cars. The power of this metaprinciple is breathtaking. Every new law that is proposed must be tested against it. If the new law passes the test (if the law is the same in every inertial reference frame), then the law has some hope of describing the behavior of our Universe. If it fails the test, then it has no hope, Einstein asserted; it must be rejected.

All of our experience in the nearly 100 years since 1905 suggests that Einstein was right. All new laws that have been successful in describing the real Universe have turned out to obey Einstein's principle of relativity. This metaprinciple has become enshrined as a governor of physical law.

In May 1905, once his discussion with Michele Angelo Besso had broken his mental block and enabled him to abandon absolute time and space, Einstein needed only a few weeks of thinking and calculating to formulate his new foundation for physics, and to deduce its consequences for the nature of space, time, electromagnetism, and the behaviors of high-speed objects. Two of the consequences were spectacular: mass can be converted